



Evolutionary Computing

HW 5

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Introduction

This detailed exposition presents an in-depth analysis and implementation of an enhanced version of the Non-dominated Sorting Genetic Algorithm II (NSGA-II), termed Fast-NSGA-II. The primary focus is on tackling the Walking Fish Group (WFG) suite of benchmark problems. These benchmarks are specifically designed for testing the capabilities of multi-objective optimization algorithms under various challenging scenarios that mimic real-world decision-making landscapes. The Fast-NSGA-II is engineered not merely as an algorithmic tool, but as a sophisticated instrument to dissect and enhance the nuances of genetic operations, particularly in adaptive crossover techniques and diversity management strategies. Our objective is a dual pursuit: to push the boundaries of algorithmic performance in terms of speed and efficiency, and to explore the robustness of the solutions provided by the adaptive mechanisms integrated into the evolutionary process.

Implementation

The WFGEvaluator Class: A Closer Look

This class is meticulously designed to encapsulate and streamline interactions with the WFG problems:

Initialization: Accepts parameters such as `problem_name`, `n_var` (number of decision variables), and `n_obj` (number of objectives). It then initializes the specified WFG problem using these parameters, setting the stage for the optimization tasks.

Solution Evaluation: Implements a method to evaluate individual solution vectors (x), computing their respective objective values based on the WFG problem's inherent functions.

True Pareto Front Access: Provides an interface to access the true Pareto front, where available, facilitating a direct comparison between the algorithm-generated solutions and the optimal fronts.

Decision Variable Bounds and Reference Points: Outputs the lower and upper bounds for decision variables, crucial for initializing populations within valid ranges and guiding mutation operations. It also provides ideal and nadir points necessary for certain performance metrics.

The FastNSGAI Class: Advanced Genetic Operations

The core of our modified NSGA-II algorithm includes several sophisticated features aimed at enhancing genetic diversity and adaptation:

Adaptive Crossover Operators: Integrates multiple crossover methods, each with initially equal probability. These probabilities are dynamically adjusted based on the relative success of each operator in contributing to the generation of promising offspring. This adaptive approach allows the algorithm to tailor its genetic operations to the problem landscape dynamically.

Population Management: Begins with the generation of a population uniformly distributed within the decision variable bounds, ensuring a diverse genetic pool. This is critical for exploring a wide search space from the outset.

Diversity Injection Mechanism: Monitors the algorithm's progress using the IGD metric; if improvements stall, the mechanism introduces new, randomly generated solutions to the population. This strategy helps mitigate the risks of converging prematurely to local optima, fostering continued exploration and enhancement of solution quality.

Detailed Genetic Operations: The algorithm meticulously implements selection, crossover, and mutation processes tailored for real-valued decision variables. These processes are designed to strike a balance between exploring new solution spaces and exploiting known good solutions to refine the search towards optimal fronts.

NSGA2Logger Class: Tracking and Visualization

To capture the evolutionary trajectory and effectiveness of the algorithm, the NSGA2Logger class is employed:

Performance Metrics Tracking: It logs the IGD and HV metrics at each generation, providing insights into how closely the algorithm-generated solutions approximate the true Pareto front and the extent of the objective space they cover.

Adaptive Crossover Probabilities: Keeps a record of the dynamic adjustments in the probabilities of each crossover operator, offering a window into the algorithm's adaptive strategies over time.

Objective Value Logging: Tracks the best objective values found across generations, presenting a clear view of the algorithm's optimization progress.

Detailed Execution and Evaluation Process

The main function orchestrates the execution of the Fast-NSGA-II across different WFG problems by systematically performing the following steps:

Parameter Configuration: Sets up the population size, number of generations, crossover and mutation rates, and other algorithmic parameters critical for defining the behavior of the genetic operations.

Optimization Execution: Runs the algorithm for the specified number of generations. During this phase, genetic operations are applied to evolve the population, while the adaptive mechanism for crossover probabilities is leveraged based on ongoing performance feedback.

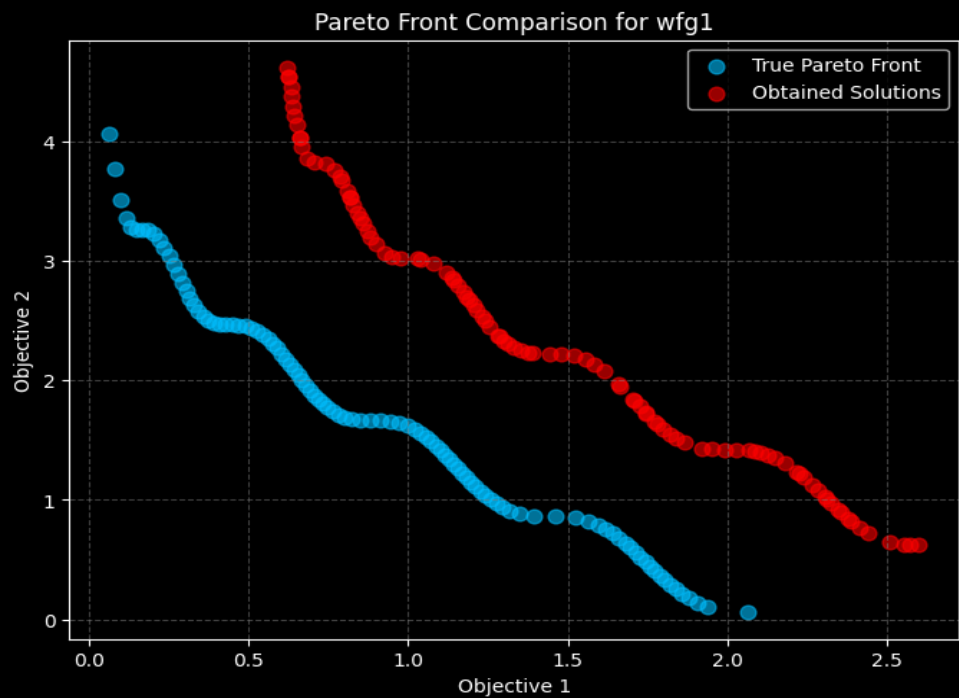
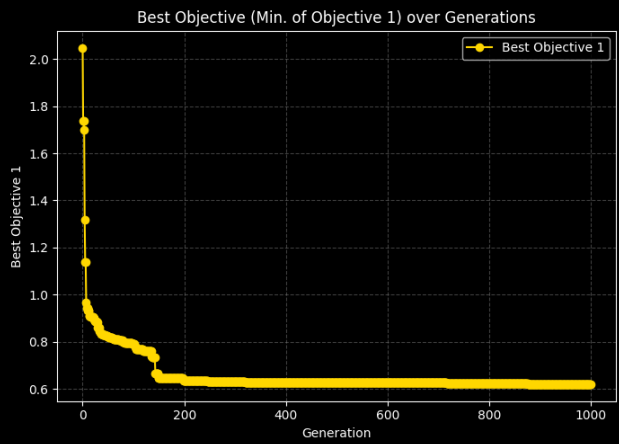
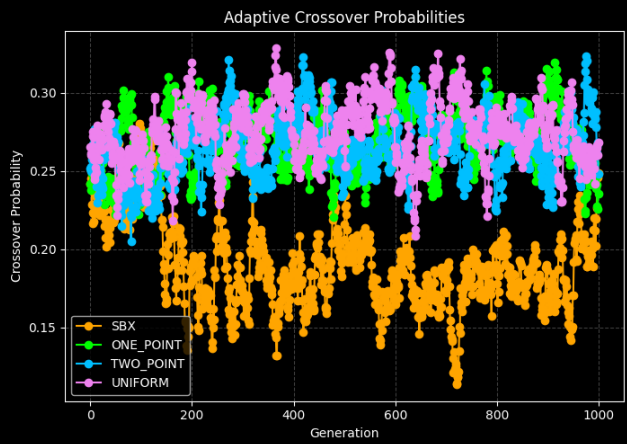
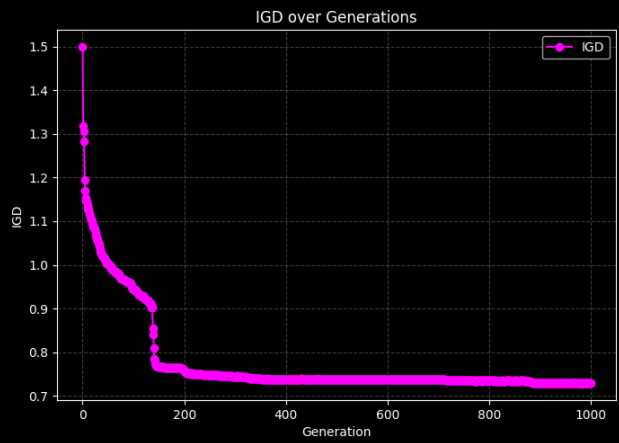
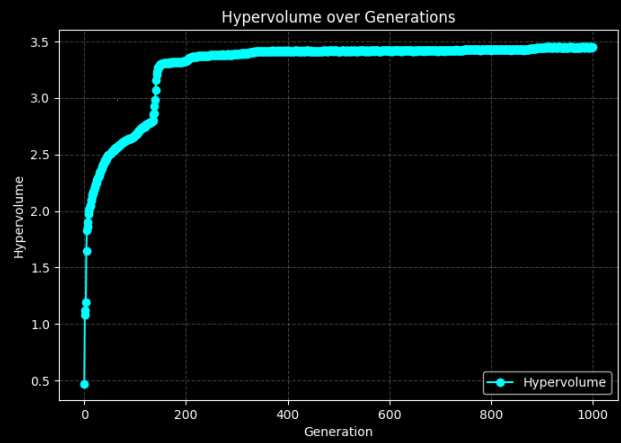
Performance Evaluation: Upon completion of the optimization cycles, the IGD and HV metrics are computed to assess the quality of the solutions obtained, measuring both the closeness to the true Pareto front and the comprehensiveness of the coverage across the objective space.

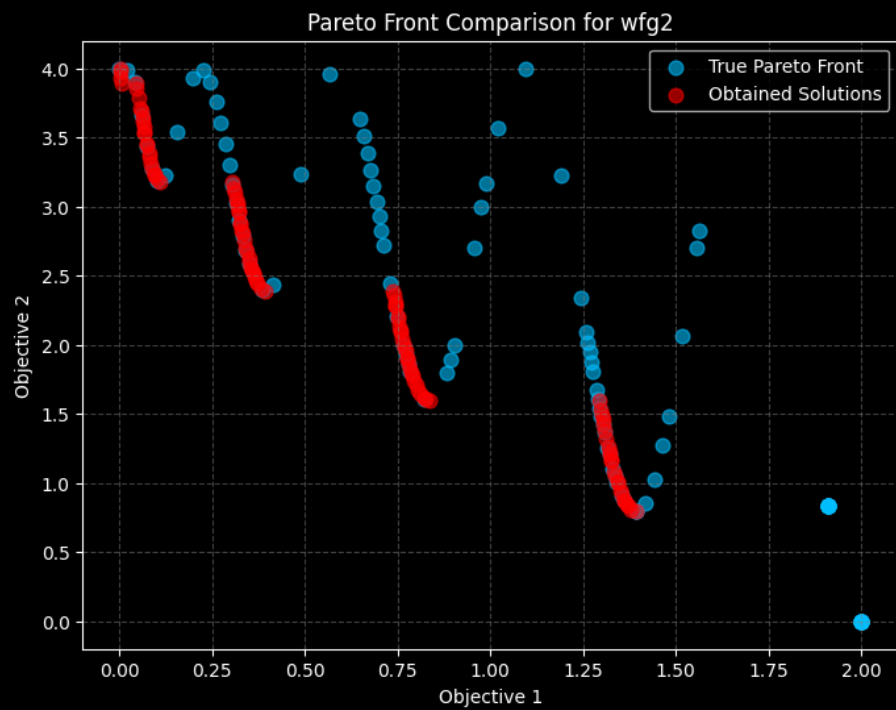
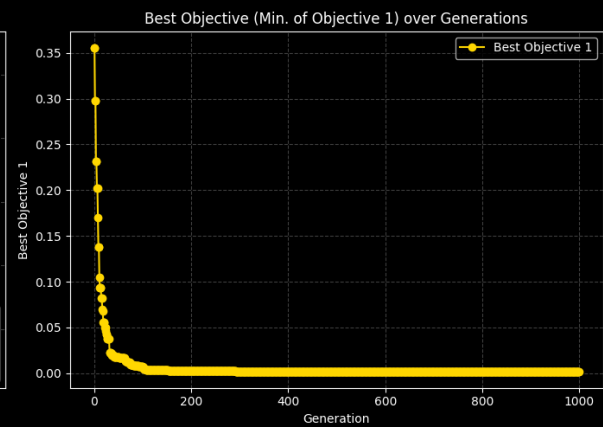
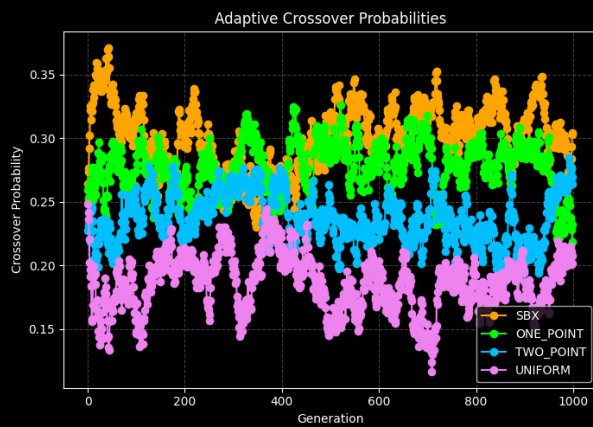
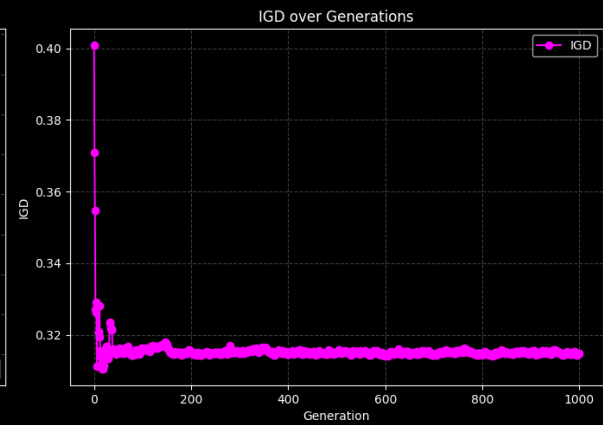
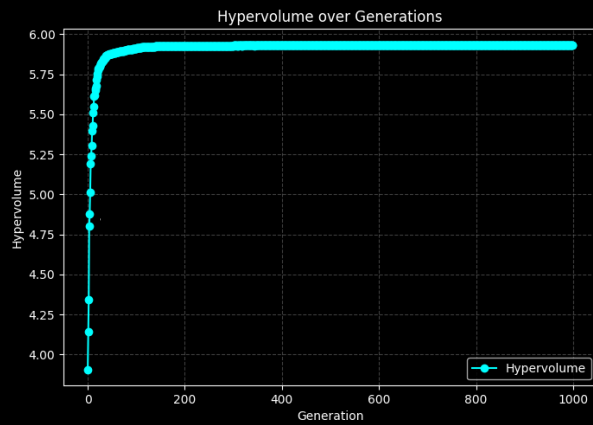
Results Presentation: Numerical and graphical results are presented, including detailed comparisons of the algorithm-generated Pareto fronts against the true fronts, showcasing the effectiveness of the algorithm in visual form.

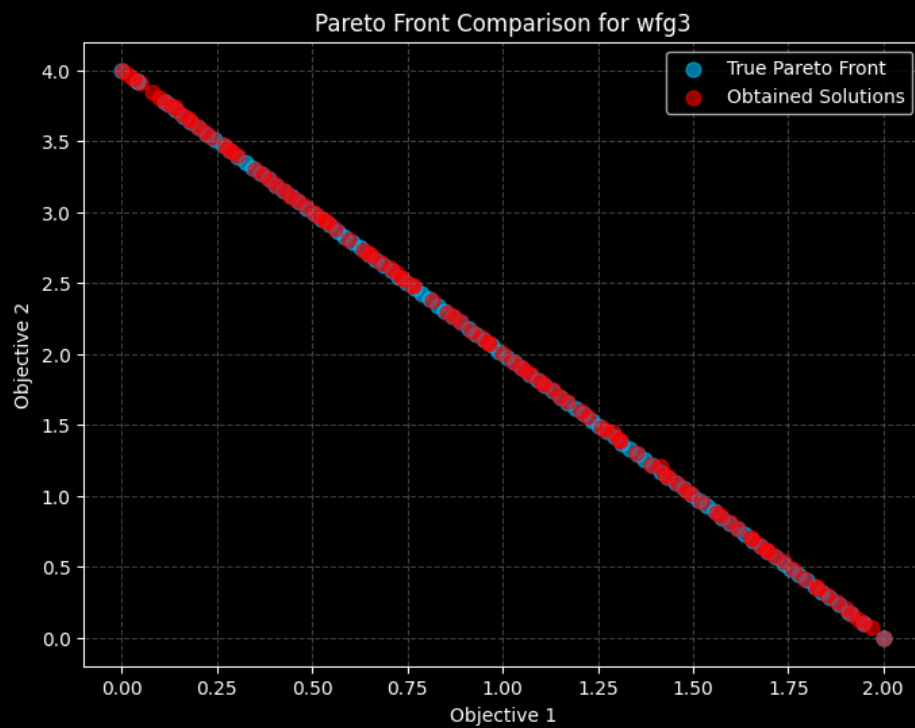
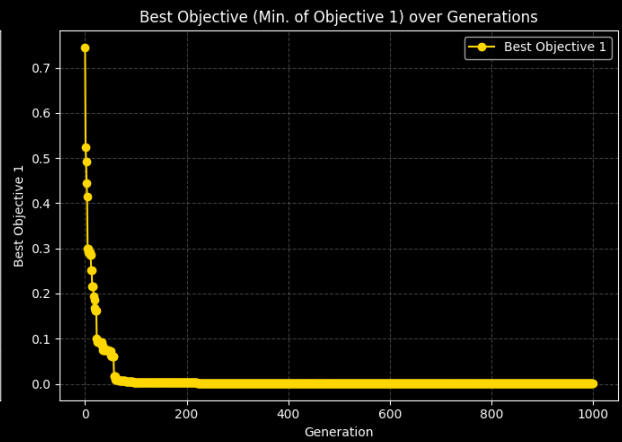
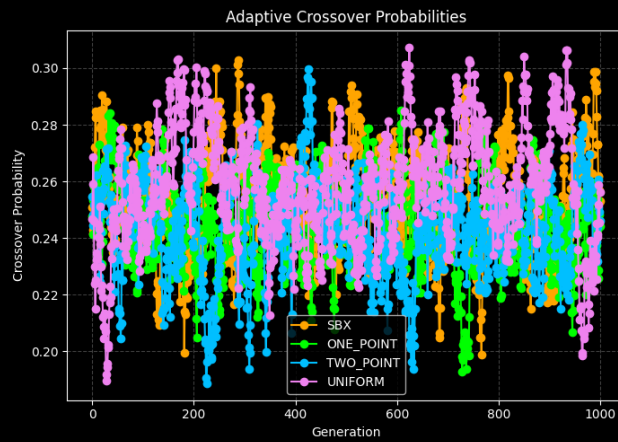
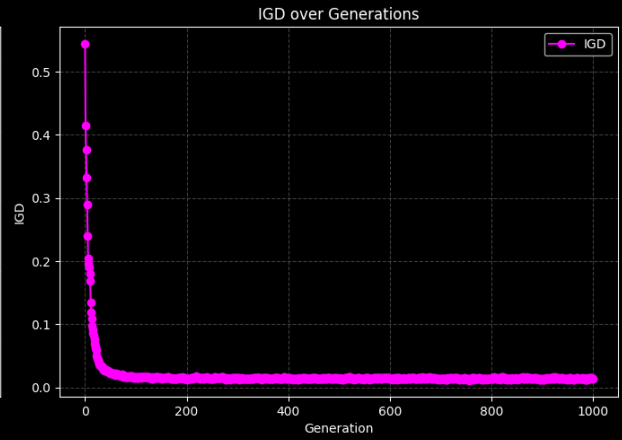
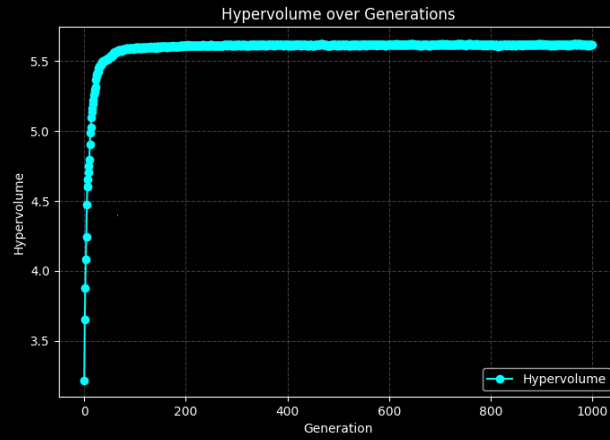
Conclusion

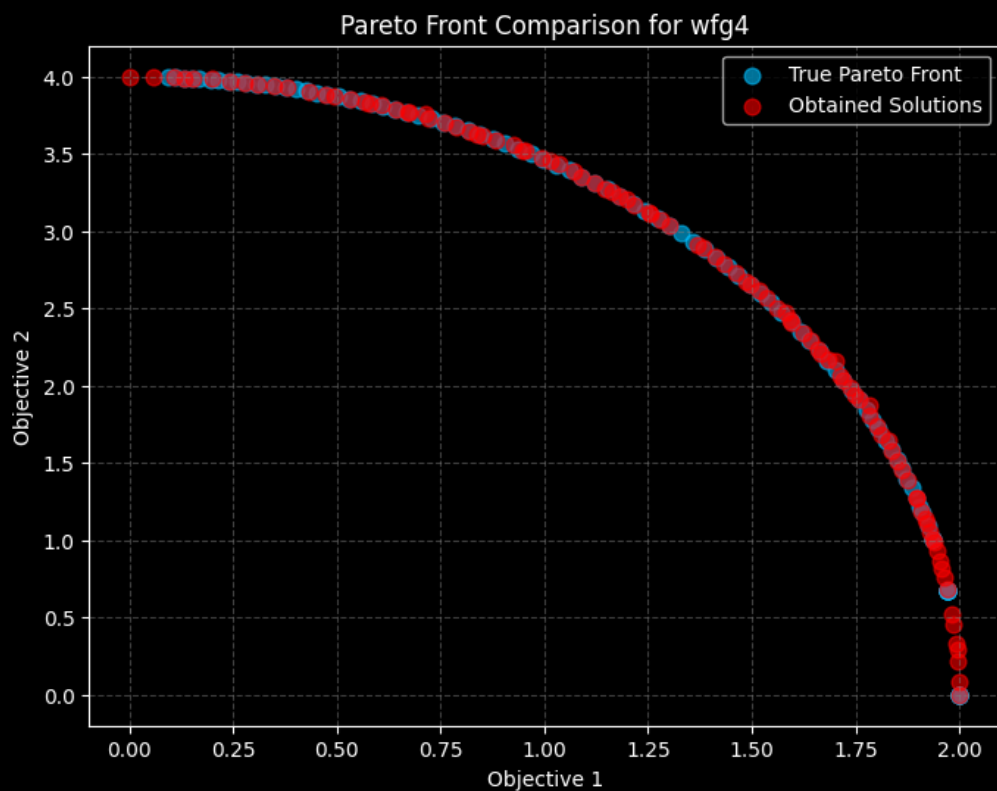
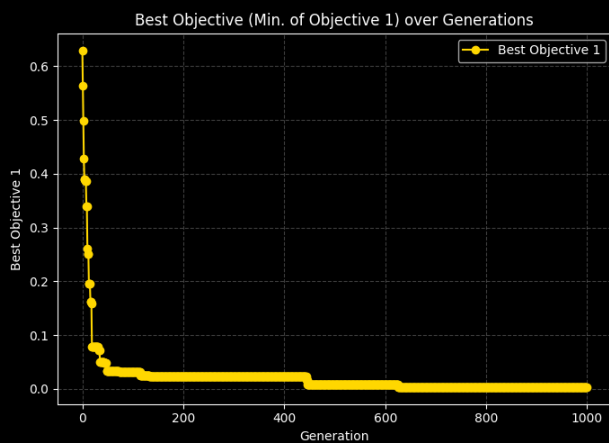
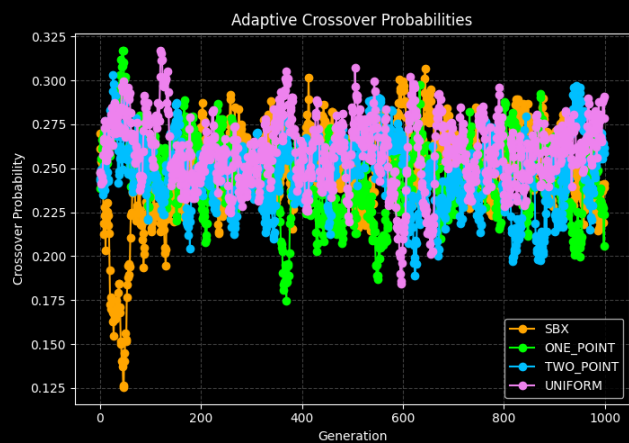
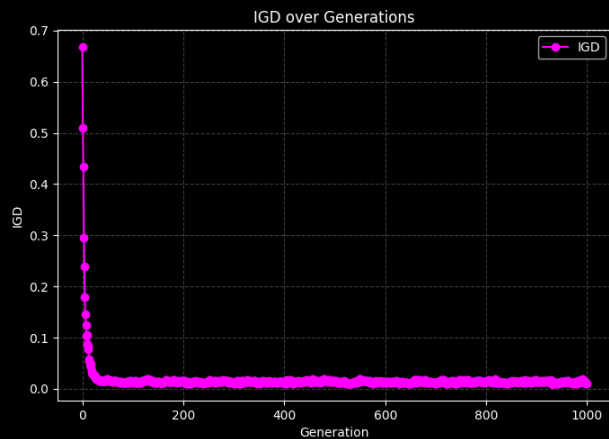
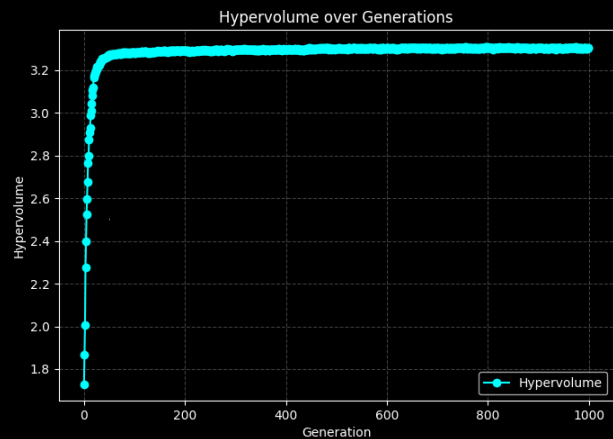
The extensive implementation and rigorous evaluation of Fast-NSGA-II within this framework demonstrate its capability to effectively address complex multi-objective optimization problems. The adaptive genetic operations, combined with strategic diversity management, not only enhance the algorithm's performance but also provide deep insights into the dynamics of evolutionary search processes. This approach not only advances the field of evolutionary computation but also sets a precedent for future research to explore more nuanced and context-aware genetic mechanisms.

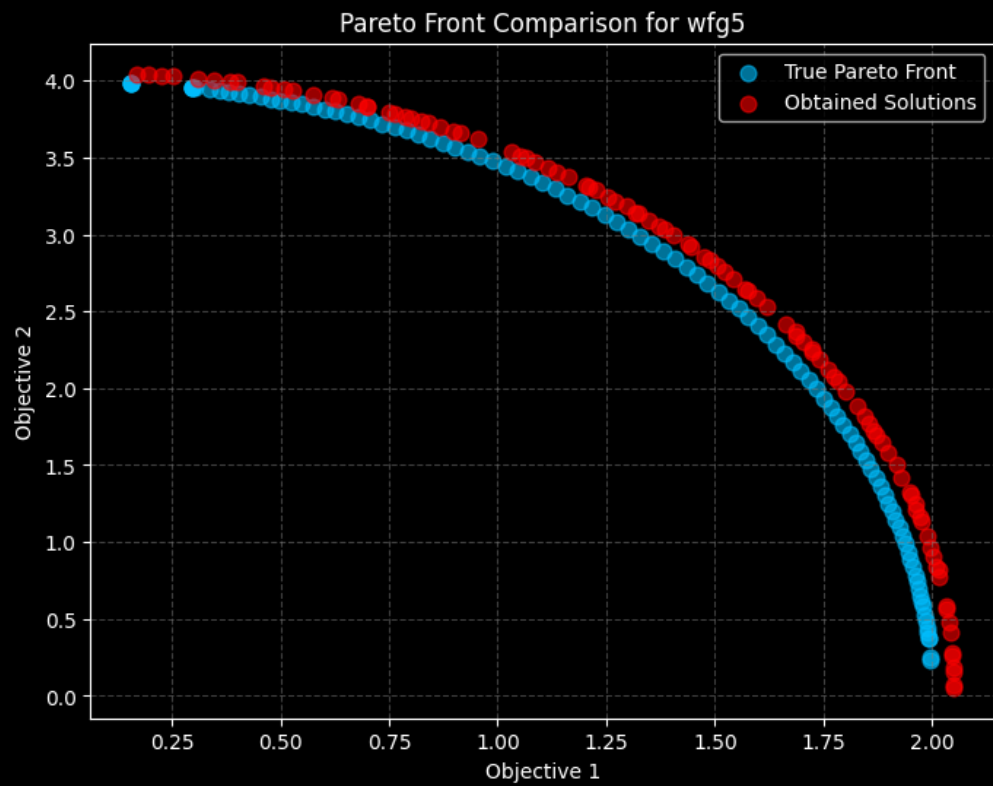
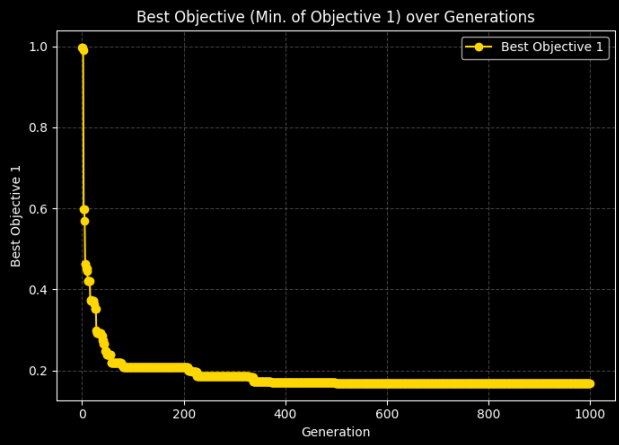
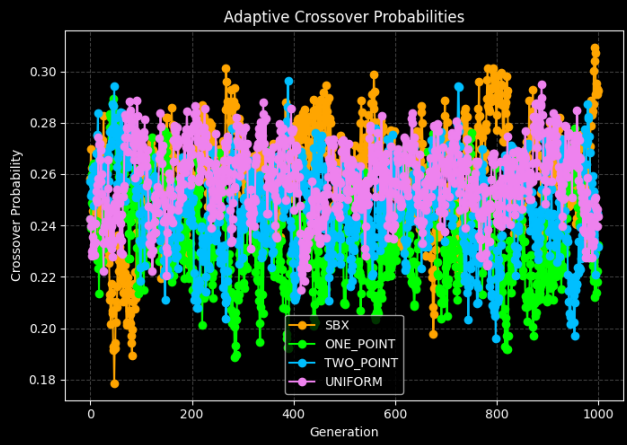
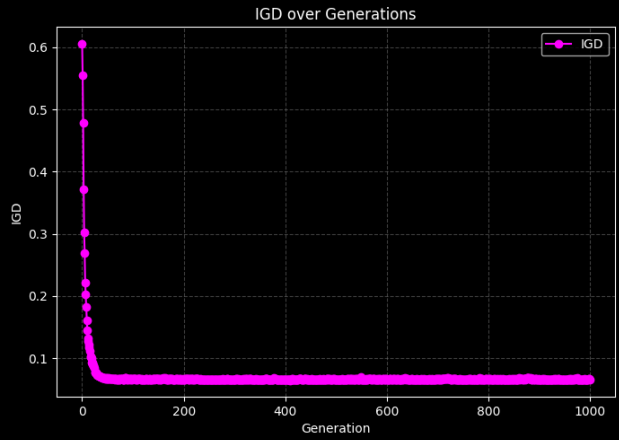
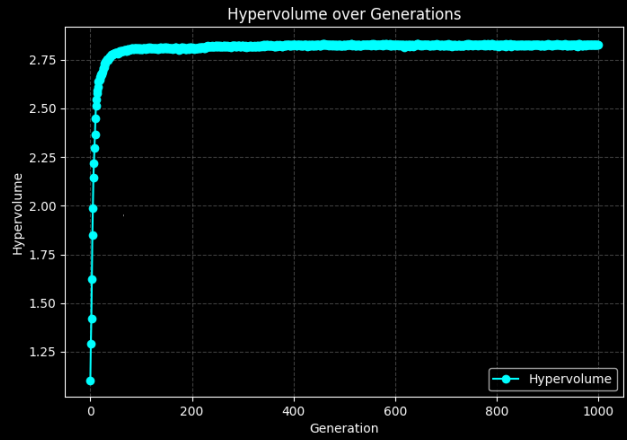
Plots

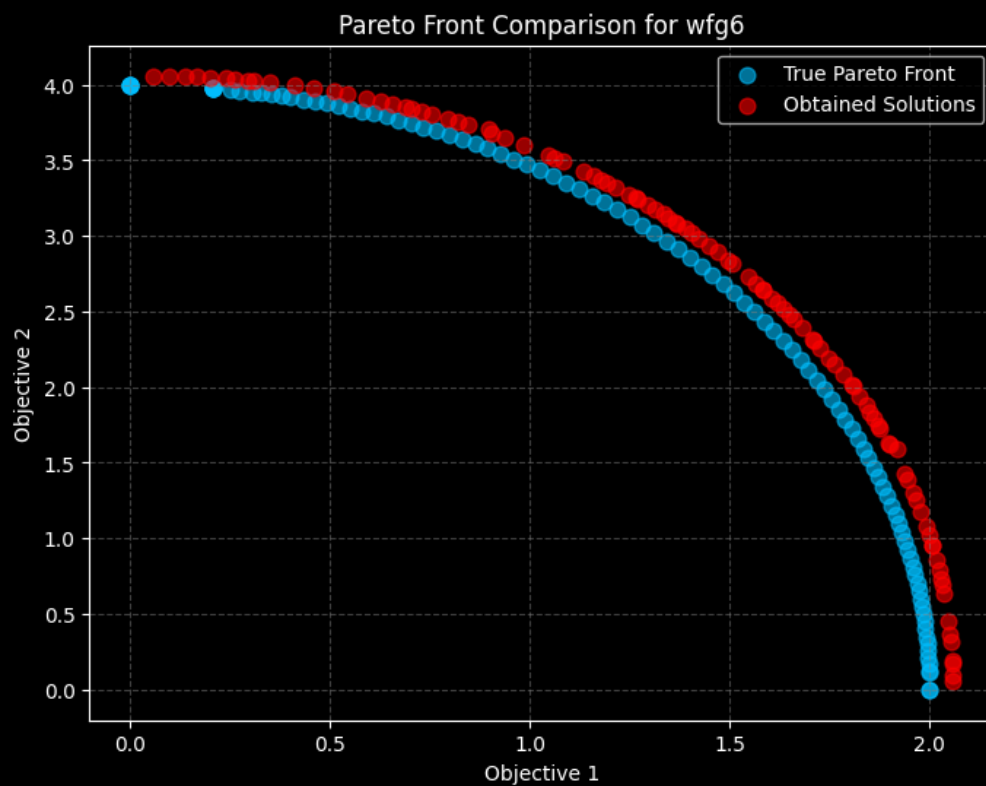
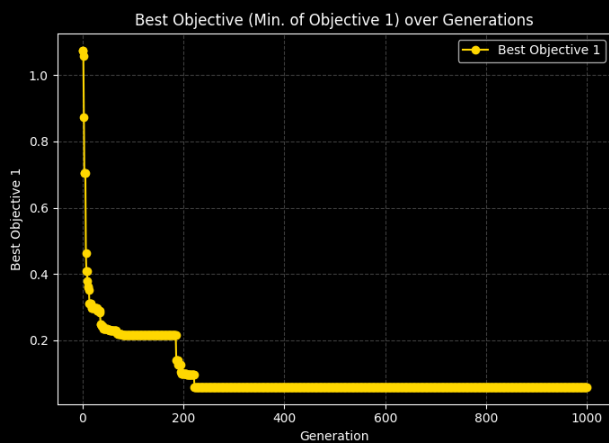
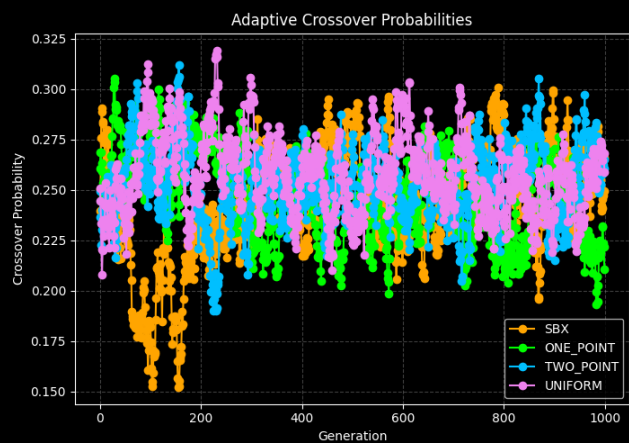
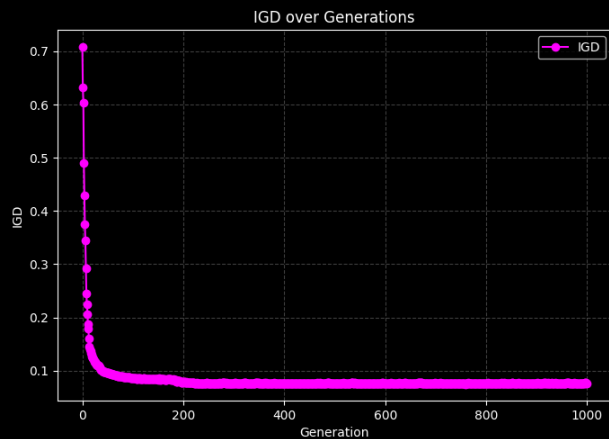
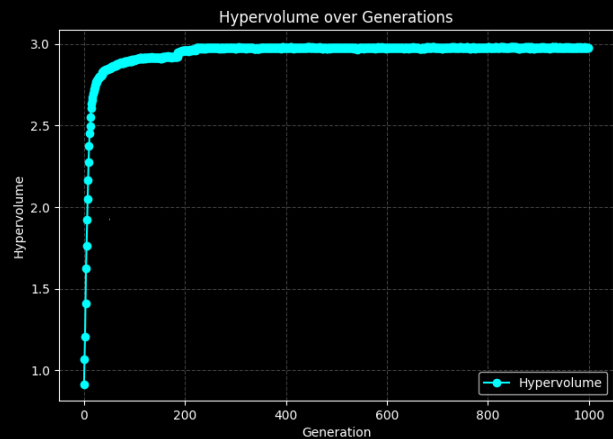


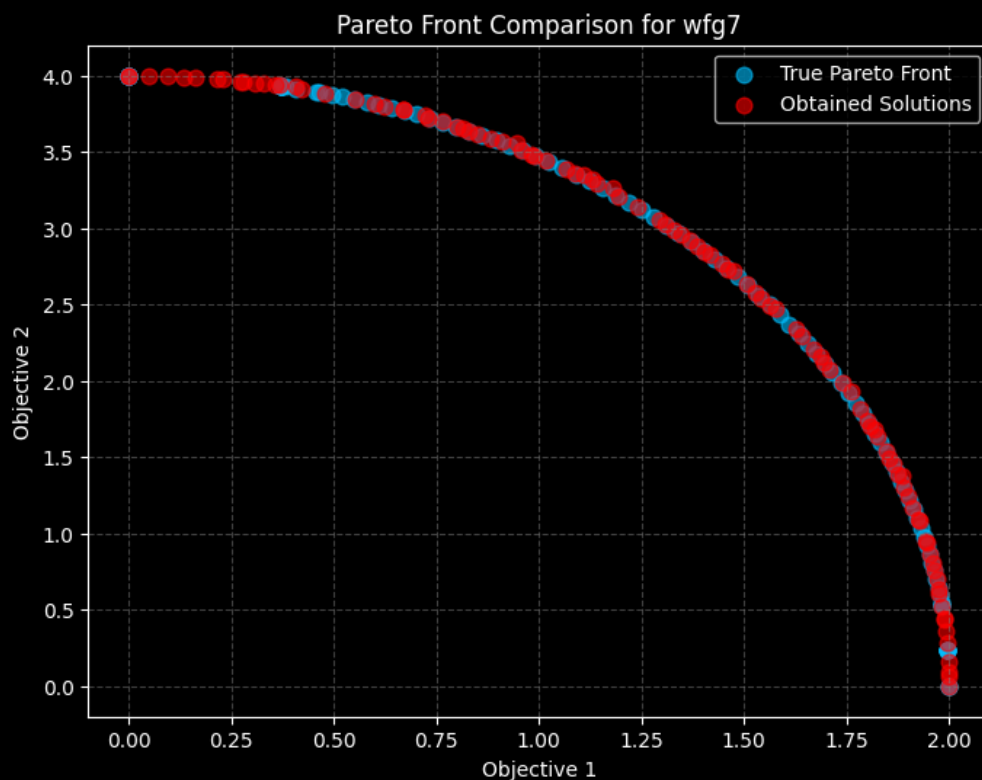
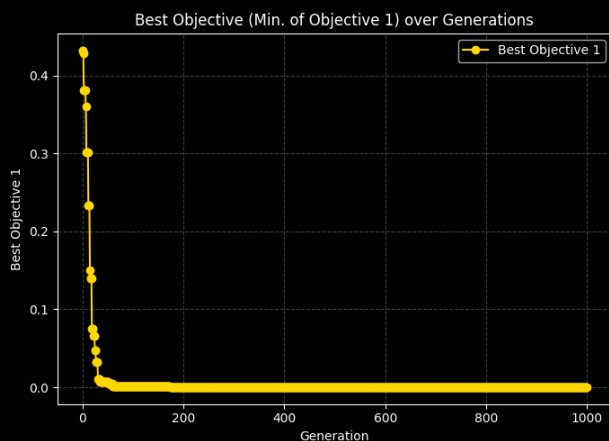
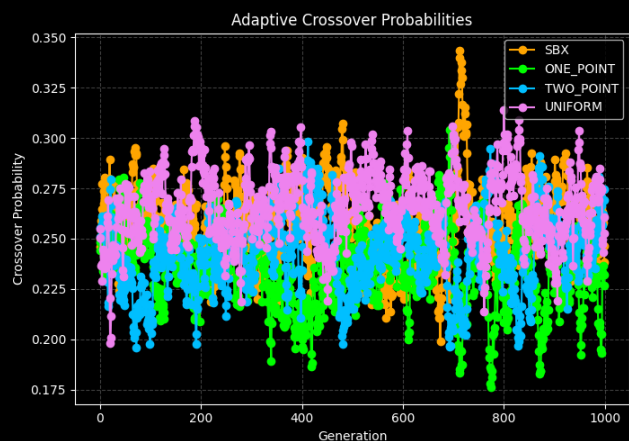
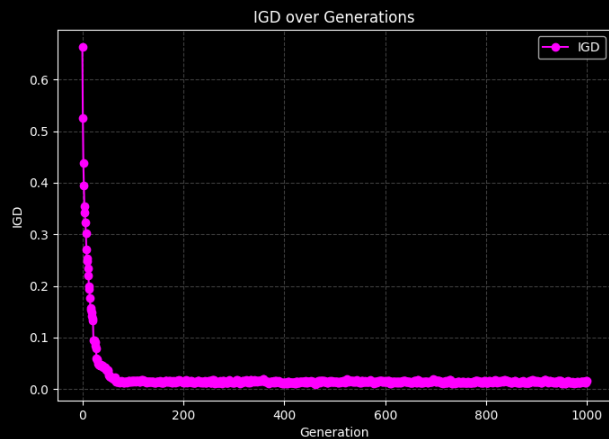
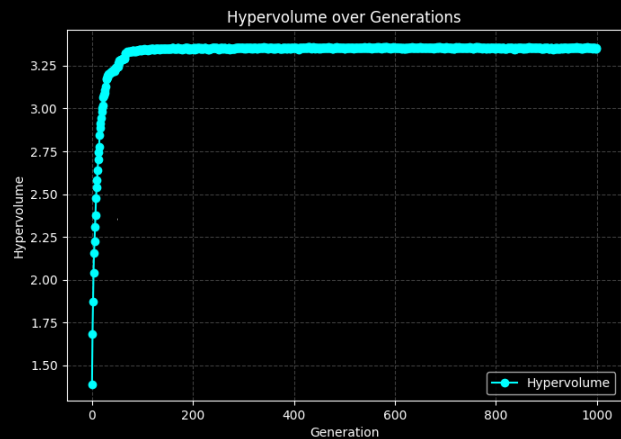


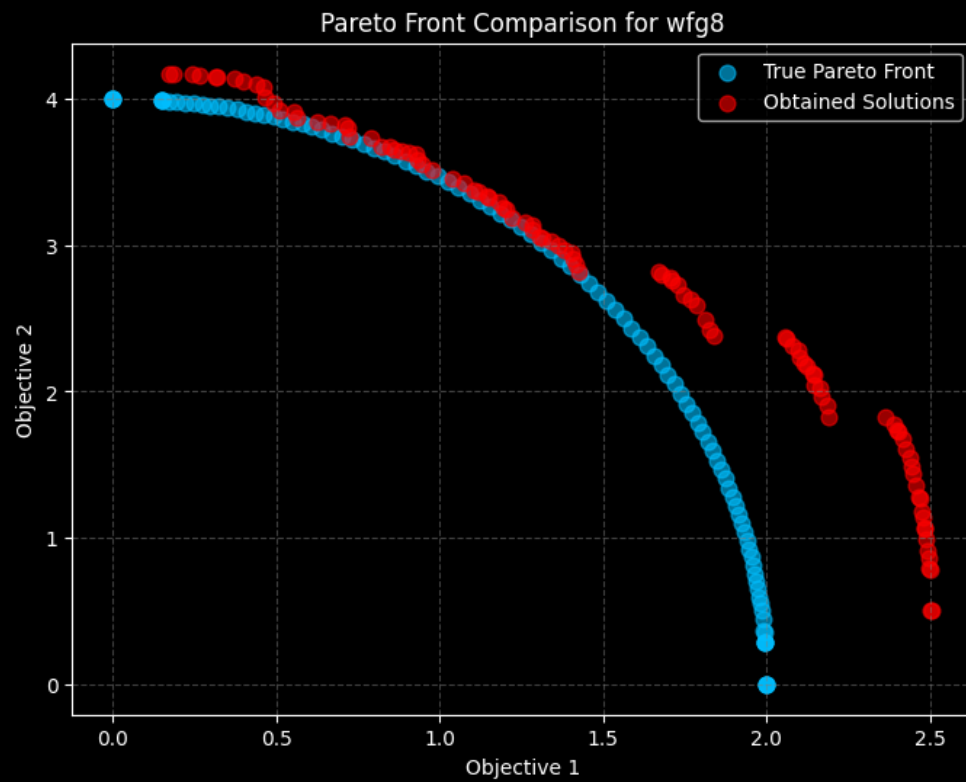
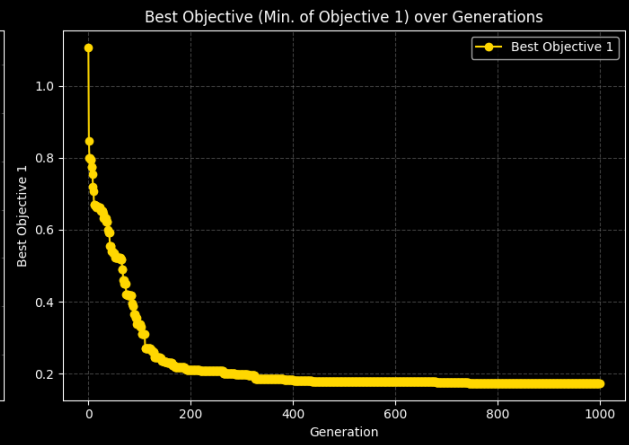
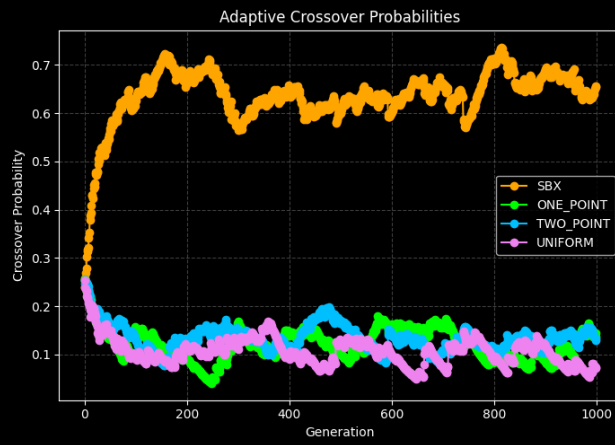
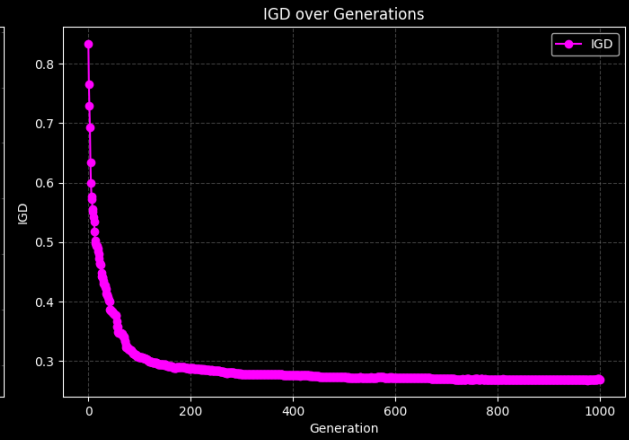
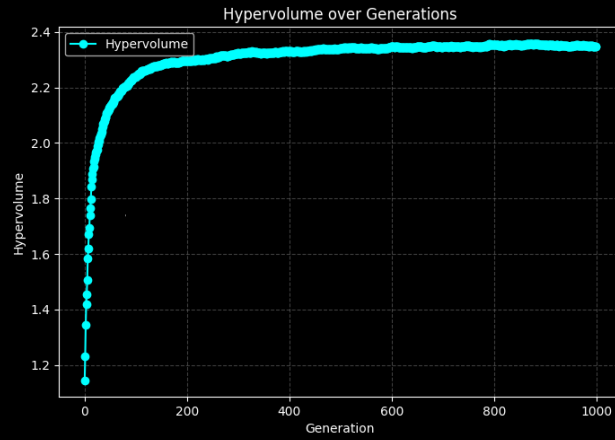


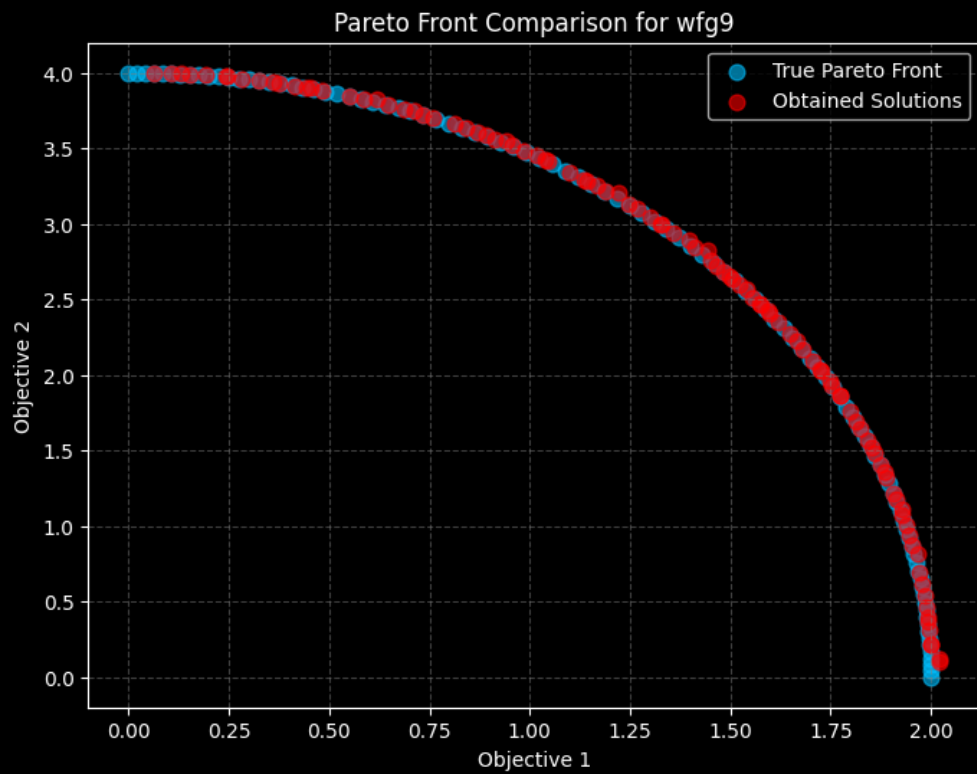
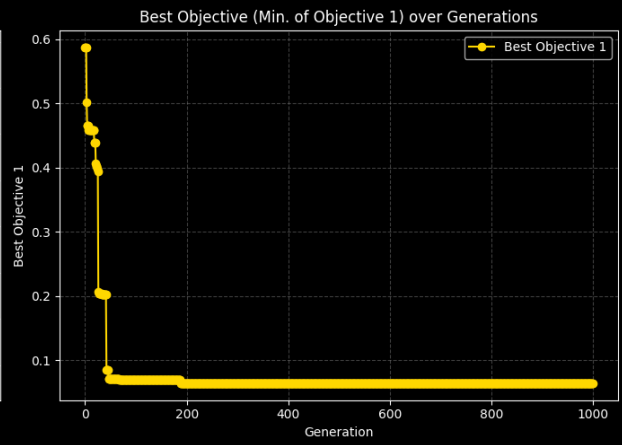
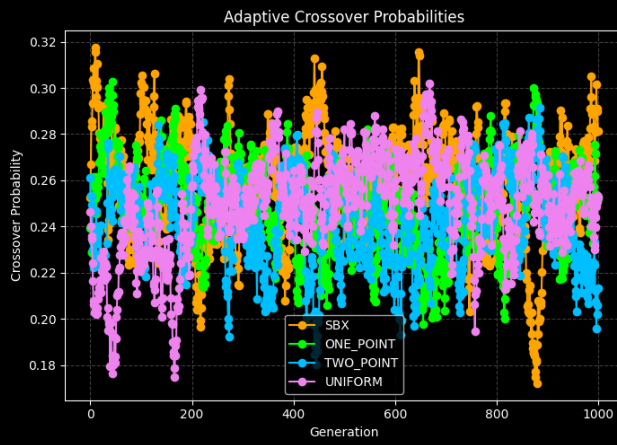
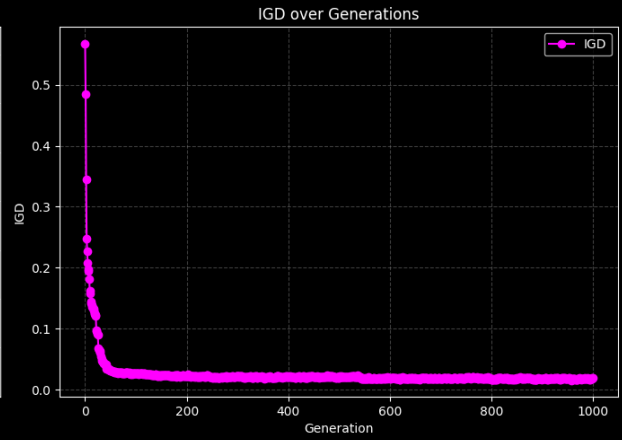
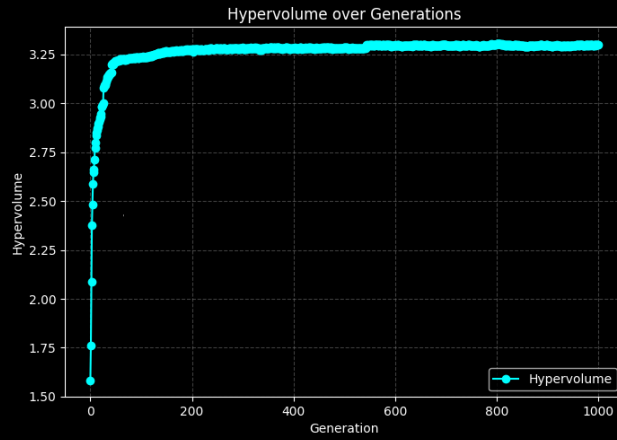












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